

When AI Tells Their Story: Researchers' Reactions to AI-Generated Podcasts as a Tool for Communicating Research

Quan Connie Gu*
School of Information
University of California, Berkeley
Berkeley, California, USA
conniequ@berkeley.edu

Daniel Hickey*
School of Information
University of California, Berkeley
Berkeley, California, USA
dan_hickey@berkeley.edu

Kimiko Ryokai
School of Information
University of California, Berkeley
Berkeley, California, USA
kimiko@berkeley.edu

Abstract

Podcasts have been recognized as an accessible and engaging medium for education and science communication. Recent advances in generative AI have led to the development of tools that transform science communication by summarizing complex research for broader audiences. However, little research has explored original authors' perspectives and reactions to these tools, leaving gaps in understanding their views on accurate representation of their work. Through interviews with 10 authors from 9 different disciplines, our study examines authors' perspectives on the accuracy, effectiveness, and role of AI-generated podcasts in engaging both academic and general audiences. The study also explores their reflections on the opportunities and risks these tools present, both for broader audiences and for authors themselves, along with their ideas for improving the design. By centering authors' perspectives, this study aims to provide insights for designing AI-assisted science communication systems that respect the integrity of academic work while enhancing accessibility.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

science communication, generative AI, podcast

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1 Introduction

Widespread access to academic literature enhances the public's scientific literacy, enabling non-experts to make more informed decisions on matters such as which policies to vote on or how to

properly treat illnesses [23]. However, the highly technical language of academic research papers presents a barrier to them being accessed by laypeople [18], and not every research paper will be summarized for a non-technical audience by expert science communicators, a problem potentially exacerbated by growing rates of academic publishing [7]. For this reason, scholars have devoted considerable efforts to develop technology that automatically summarizes research papers and scientific concepts in more accessible ways such that they can be understood by a variety of audiences [4–6, 10, 12, 14, 19]. While the current literature on AI-generated summaries of research mostly focuses on written text summaries, researchers have explored the use of automatically generated podcasts to summarize news articles [17] and textbook chapters [11]. Prior research shows that podcasts are an entertaining and accessible way of educating medical students that do not compromise the instructional quality of traditional learning methods [16]. For this reason, AI-generated podcasts describing research papers show great promise for increasing accessibility to science and improving scientific literacy. However, scholars have previously expressed concern that generative AI systems can “hallucinate,” or generate explanations that are unfaithful to the source material they intend to summarize [1, 2]. These concerns add to existing work demonstrating the worrying prevalence of misinformation about science in society, which predate the current boom in AI-generated content in the media [9, 21]. Given concerns about the veracity of AI-generated content, it is important to understand how faithfully authors of research papers think the current state of the art in AI-generated podcasts interpret their work, and what consequences such tools have for the current landscape of science communication. For this reason, we present an analysis of 10 interviews with authors, seeking to understand their perspectives on how faithfully AI-generated podcasts represent their work and how useful such podcasts can be in presenting their work to audiences outside of their field. Our findings show that hallucinations are common, authors have mixed perspectives on whether such tools should be used to communicate to broader audiences. We show how the podcasts prompted authors to reflect on the way they produced their work, and we summarize insights authors had into improving the design of AI-assisted science communication tools. We contribute a fresh perspective on the societal implications of AI-assisted science communication. To our knowledge, this study is the first to qualitatively survey authors' perspectives on the representation of their work by AI podcasts, and we contribute insights into how researchers can inform the design and evaluation processes of AI-assisted science communication tools.

*These authors contributed equally to this work

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2 Related Works

There is a substantial body of research dedicated to making academic research papers more accessible to audiences outside of academia. Long before the current paradigm in AI-generated media, certain academic publishers have included “plain language summaries” along with each paper they publish as an effort to make academic research more accessible to the general public [18, 20]. Plain language summaries are often short descriptions of a given paper that describe the significance of the paper in a way that is free of technical jargon. Recent advances in generative AI allow plain language summaries to be more efficiently generated for academic research, even when the publishers of such research do not adopt the practice of writing plain language summaries [10, 12]. Researchers have also experimented with AI-assisted science communication that extends beyond simple plain language summaries. For example, “Paper Plain” is an interactive tool that allows users to view automatically generated summaries of specific parts of research papers [6], and Shin et al. generate “design cards” for academic research papers, which incorporate visual elements related to the research along with a written summary. Another promise of AI-generated science communication is personalization – through LLM prompting, descriptions of scientific concepts can be easily adjusted to fit with the level of background knowledge of the intended audience [4]. Similarly, Do et al. 2024 find that tailoring educational podcasts to the hobbies and interests of the listener improves learning outcomes [11]. AI-generated science communication is evaluated across a variety of methods, including computational methods [13] or usability studies that include surveys and interviews [6, 19]. Common concepts used to evaluate plain language summaries are informativeness, faithfulness, simplification, and coherence [13]. However, scholars have also developed methods to measure other important attributes of science communication, such as the science communication strategies of storytelling and use of analogy [3] or “distortions” such as changes in sensationalism or certainty levels in pieces of science communication when compared to the original source material [22]. We build on prior research in AI-assisted science communication by evaluating AI-generated podcasts as a medium for conveying complex research. By focusing on authors’ perspectives, we examine the societal implications of disseminating outputs from such tools and explore how they influence researchers’ views of their own work. We hope to inform the design and evaluation of AI-assisted science communication systems that uphold the integrity of academic research while improving accessibility.

3 Methodology

We conducted 10 semi-structured interviews with authors of research papers to explore their perceptions of an AI-generated podcast based on their academic publications. Prior to each interview, we asked each participant to select a recently published paper of theirs that they played a key role in (e.g., PI or first author). We used these criteria to ensure that authors were being interviewed for the work that was most relevant to them. Each session began with a 3-5 minute introduction, where participants provided a summary of the paper designed for an audience of scholars outside their field. This introduction served as an opportunity for researchers to highlight key points in their work, refresh their familiarity with the

content, and practice communicating to an audience outside their field. Then, to provide context for their reactions to the AI podcasts, we asked authors background questions about their podcast listening habits and their familiarity with generative AI. Following this, we played a podcast generated to summarize the paper they described using the “audio overview” feature of Google’s NotebookLM tool¹. The podcasts lasted 14 minutes on average, with a minimum time of 10 minutes and a maximum time of 18 minutes. After playing the podcasts for the authors, we structured the discussion around five predefined themes: Overall Impressions, Utility for Broader Communication, Trust and Accuracy, Design Implications, and Reflections on the Process. These themes were collaboratively developed by two researchers based on feedback from three pilot interviews and were designed to guide the discussion toward understanding both the practical and conceptual implications of AI-generated podcasts. This particular set of themes was chosen to explore participants’ views, focusing on the quality of representation, ethical considerations, and how these tools might inform or challenge their own practices as researchers. We aimed to capture a broad range of insights relevant to both science communication and the design of AI systems.

3.1 Participants

We recruited 10 participants (6 male, 2 female, 1 non-binary, 1 not reported), including 2 faculty members, 7 PhD students, and 1 master student, representing 9 different departments within the university’s research community. To sign up, participants completed a screening form to confirm their eligibility as researchers with published papers in their field. The form also collected relevant information, including their field of study, academic position, and a publicly accessible paper they wished to discuss during the interview. Participants were selected based on their eligibility and to ensure diversity across fields of study. After confirming eligibility, we contacted participants via email to schedule their interviews. All interviews were conducted over a video conferencing platform. Informed consent was obtained, and identifying features such as names and positions were removed to protect the anonymity of the participants.

Table 1: Participants’ Positions and Departments

ID	Position	Department
P1	PhD student	EECS (Computer Science)
P2	Faculty	Landscape Architecture & Environmental Planning
P3	PhD student	EECS (Electrical Engineering)
P4	PhD student	Chemistry
P5	PhD student	Astronomy
P6	Faculty	Education/Cognitive Sciences
P7	PhD student	Mechanical Engineering
P8	PhD student	Education
P9	Master student	Master of Design
P10	PhD student	Information Science

¹notebooklm.google.com

3.2 Analysis

Each interview lasted approximately 1 hour. This paper focuses on the qualitative analysis of the semi-structured interview questions asked after playing the podcast. The data collected during the interviews was transcribed, cleaned, and prepared for analysis. To analyze the data, we applied an inductive approach to coding [8]. Two coders independently developed initial codebooks based on granular codes. These initial codebooks were then reviewed collaboratively by both coders, who worked together to finalize a unified codebook after jointly reviewing two transcripts. Using the finalized codebook, we identified recurring themes across the transcripts, and relevant codes were then grouped.

4 Findings

In this study, we investigate researchers' perspectives on AI-generated podcasts, focusing on their accuracy in representing academic work, their utility for science communication across diverse audiences, and their influence on how researchers reflect on their own work. We also explore how researchers envision future design possibilities for AI-generated podcasts. Our findings highlight both the benefits AI podcasts offer and the challenges they pose for communicating science to academics and the broader public.

4.1 Trust and Accuracy

We began by asking participants questions about the accuracy of the podcast, including what stood out to them and how well their research was represented. We then extended the conversation to explore their trust in AI podcasts to provide accurate information about their research. Our goal was to understand, from the researchers' perspective, how they evaluate the reliability of such AI-generated outputs and the factors influencing their trust or skepticism.

4.1.1 Hallucination. Hallucinations identified by participants highlight the challenges of AI tools' reliance on unrelated data sources and lack of field-specific contextualization. Some inaccuracies arose from external assumptions or generalizations not rooted in the paper itself (P2,3,4,6,8,10). P3, a PhD student in electrical engineering, noted that the podcast inaccurately emphasized affordability, a point unrelated to their work. They explained that while the materials needed for their research are often described as inexpensive online, this was not reflected in their research or experience, stating, *"I ordered the parts for this, I know that it was not cheap."* Other hallucinations were attributed to misrepresentation due to lack of understanding of the field (P1,5). P5, a PhD student in astronomy mentioned the podcast *"confused what color in an astronomical sense is vs. our everyday experience."* In some cases, the podcasts overreached in their attempt to emphasize real-world implications, leading to overhyped conclusions, participants described them as *"little bit far-fetched."* (P1), *"misleading"* (P4), and *"too much exaggeration"* (P6).

4.1.2 Plagiarism. Participants expressed concerns about potential plagiarism in the AI-generated podcasts, noting instances where the content directly replicated their work without proper attribution. P5 described *"That is an analogy that I specifically used in my*

presentations, and I wasn't aware of it being used anywhere else... It's a little bit creepy to know that it pulled that from my slides." P7 shared similar concerns, *"It took some stuff from articles I wrote and copied it word for word. It was scarily similar."* and *"it was pulling word-for-word from things I've said."* Additionally, P2 highlighted the misattribution of findings, explaining, *"When it pointed out that groundwater could affect twice as much land, we say that, but we clearly cite someone else's paper. That's not our finding."* These instances raised questions about the AI's ability to distinguish between general knowledge and proprietary content, as well as the ethical implications of reusing researchers' work without acknowledgment.

4.1.3 Lack of Field-Specific Knowledge. Participants highlighted that the lack of field-specific background knowledge in the AI podcasts led to confusion about the roles or voices of the podcast hosts, particularly whether they were meant to sound like outsiders or experts (P3,6,7). Concerns include inconsistencies of field-specific jargon, with P5 noting, *"It's kind of weird that they use some terminology, [...] I recognize the terminology but then they change it"*, and the shortening of jargon in ways unfamiliar to the field, with P3 stating, *"There was just a few shortenings [...] If I were to hear that [...] I'd be like, I don't really know what they're talking about"*. Other participants pointed to failures in adhering to disciplinary standard practices, with P2, a professor of architecture, observing that the podcast did not convert units from hectares to square miles, *"That's not normal speech [...] we normally would do [the conversions]."*

4.2 AI Podcast for Science Communication

We asked participants to reflect on the potential audience for AI-generated podcasts of their academic work, considering both the opportunities and challenges these tools present. They explored how AI podcasts could shape science communication in different contexts and the roles they might play in reaching both expert and public audiences.

4.2.1 Perspectives on AI Podcasts for Experts vs. the Public. Participants reflected on the potential audience for AI-generated podcasts and the effects they might have. Most highlighted their potential as tools for public science communication, describing them as an effective way to make academic research more accessible (P1,4,5,7,8). They suggested audiences such as teachers, people with an interest in science, and middle and high school students. P4 remarked, *"I think the AI really suggested a good picture, and I think I would almost keep what it has presented [for the general public],"* while P6 appreciated the focus on simplicity, noting *"I like that [the podcast] honed in on the key big ones."* However, some researchers raised concerns about the potential for spreading misinformation (P1,3,5,6), with P5 cautioning, *"If it's speaking to people who don't know any better, if they get some inaccuracy then that might really take hold, and that would be very bad."*

When considering the podcast's use among experts, participants were less enthusiastic. Many found the content *"too broad and big-picture to be helpful"* (P7) and *"way too conversational and skips a lot of important details"* (P1). However, some participants acknowledged potential benefits for academic audiences, such as discovering new ways of introducing research (P3), gaining an understanding of

unfamiliar concepts and areas (P4), and forming deeper impressions of certain ideas through casual listening, which they felt was more engaging than their usual practice of skimming papers (P9).

4.2.2 AI as an Assistive Tool. We then further explored the utility of AI-generated podcasts, encouraging participants to consider how such tools might be used. Participants generally viewed AI podcasts as assistive tools but emphasized the need for caution regarding their accuracy and reliability. P7 stressed the importance of treating AI tools as supplementary, stating, *“There’s always room for error... please always go back to the root work and treat this as an assistive device which could have incorrectness.”* Similarly, P4 highlighted the importance of verifying AI-generated content, explaining, *“I will probably double-check it by going through the original source.”* P6 highlighted AI podcasts’ role as a secondary source, *“You should read the primary source, not just the secondary or the third... these podcasts are a secondary resource.”* Participants also cautioned against over-reliance on AI tools, with P9 mentioning, *“Let it explain a paper to us, but don’t rely on it too much.”*

4.2.3 Challenges. Participants highlighted the challenge of balancing the specialized nature of their work with the need to make it accessible and engaging for broader audiences. First, participants noted that their papers are often highly specialized and not always representative of their field, with P5 describing their work as *“in the weeds about particular things,”* and P3 referring to as *“very specific academic paper,”* while broader audiences are typically more interested in *“understanding things on a more conceptual level for outreach”* (P5). Second, participants highlighted the extensive background knowledge required to fully grasp their work. P4 pointed out, to understand their work *“there needs to be a whole semester of quantum mechanics, and that’s something an AI podcast cannot explain in a 12-minute conversation.”*

4.3 Reflections on Research Representation

Participants discussed how AI-generated podcasts represented both their research and their authorial voice. We aim to understand how researchers perceive the alignment between the podcast’s representation and their original work, as well as the podcast’s impact on the perception of their paper.

4.3.1 Lack of Nuance and Personal Voice. Researchers highlighted that AI-generated podcasts often failed to capture the nuance and intent of their work, feeling overly standardized and lacking the authentic voice of the researcher. P8 noted, *“It misses out on a lot of nuance,”* while P7 described the structure as a *“cookie cutter format,”* particularly highlighting how the podcasts concluded with ethical considerations that felt formulaic. Participants also noted that the podcasts overly focused on background information while misrepresenting key priorities (P2, P7, P8). Additionally, researchers expressed frustration with the podcasts creating unintended voices or tones that diverged from their intent. P8 shared, *“There are a few times where they’re like, ‘this almost seems like science fiction,’ [...] But it doesn’t in the paper.”* P1 explained, *“If the podcast makes claims or has a personality that the researcher doesn’t want or is not representative, then that could also be very bad.”*

4.3.2 Critical Reflection on the Paper. Participants reflected on how listening to the AI-generated podcast influenced their views of their papers. While most indicated they would not revise their work as a result, the experience prompted considerations about clarity and audience (P2,3,4,5,6,8). P4, responding to hallucinations in the podcast, noted that an inaccurately represented section made them consider whether that part of the paper needed further clarification. Similarly, P5 shared how hearing specific phrases directly pulled from their paper led to self-reflection, stating, *“It made me double-think about certain phrases, made me think I should use better wording.”* Some participants also reflected on the implications of AI as a potential future audience for academic writing, with P8 questioning, *“Do I think people are going to upload things I write into AI? [...] Should I be in conversation with the AI as well as my field?”*

4.4 Opportunities and Imagined Design

Participants were led through a process of imagining and designing improved AI-generated podcasts. This process aimed to invite researchers to identify design opportunities that balance accessibility, adaptability, and trustworthiness, therefore informing the development of AI-assisted science communication.

4.4.1 AI Podcast as a Medium. Participants have mixed views about the modality of podcasts as a way to communicate research. Some found the dialogue format engaging (P2,3,7,6,8), with P2 highlighting the podcast with the male- and female-sounding hosts² made it *“more engaging for people from both genders to listen”*. Participants also emphasized the value of podcasts for multitasking, describing them as a practical way to stay updated on research while performing everyday tasks, particularly for people in academia (P2,5,8).

However, some find the tone inauthentic (P4,5,8), too conversational (P1), and overly excited or uncritical (P3,4, 8,6,10). P4 noted, *“Of course, I’m grateful that the AI-generated podcast can decorate my work to sound like an important, groundbreaking, highly valuable work of art, but I don’t know what the outcome of the conversation is because they’re exaggerating way too much.”* Some participants pointed out the absence of genuine human experiences and emotions that are central to traditional podcasts. P3 noted, *“[A famous podcast host] deliberately [...] talk[s] about her own experience related to the thing. And that was missing from [the AI-generated] one.”* Similarly, P10 commented on the disconnect when the AI podcast host mentioned *“my grandma,”* stating, *“That was wild because he doesn’t have a grandma.”* P10 elaborated further that the value of personal anecdotes is lost when the personal anecdotes do not come from a human.

4.4.2 Personalization and Adding Visuals. We asked participants to identify what was missing from the AI-generated podcasts and to envision their ideal version. Personalization emerged as a key suggestion, particularly for tailoring podcasts to audiences with varying levels of familiarity with the field. P4 explained, *“If there’s an option to make it a little bit more technical, I hope for people inside my field it’s a little bit more technical.”* Participants also emphasized the potential for interactivity to enhance personalization. P6 suggested the use of a flashcard feature to adapt the content based on the

²The Notebook LM podcasts always have one male-sounding and one female-sounding “host” to describe each paper.

listeners' familiarity with key terms from the papers, while P1 proposed creating a more interactive experience and making it a "virtualization of the author of the paper."

The lack of visual aids to support understanding was frequently highlighted by participants (P1,3,4,6,7,9,10). Participants noted that certain concepts, such as equations, figures, and gestures, are difficult to convey effectively in a purely audio format. P7 emphasized looking at equations and numbers and figures are "core parts of the scientific process." P10 emphasized the role of visuals in aiding recalling P10 highlighted their role in aiding recall, explaining that "seeing the shape of the curve as opposed to remembering the pattern that was being described" makes it easier to retain information. P6, an education researcher studying the role of gesture in physics classrooms, pointed out how the lack of visual cues hindered clarity when explaining dynamic tasks, "It should have been clearer that now the hands are moving up and down, but one is still moving up and down like before, but the other one is moving sideways. [But] it's a podcast, you cannot see me."

4.4.3 Ensuring Safety and Accuracy. Participants emphasized the importance of implementing safety measures, including transparency, disclaimers, and author-developer collaboration. Transparency about how the podcasts are created and their limitations are commonly mentioned (P1,3,7,8). This includes educating audiences about the tool's limitation and clearly define the podcast's role in science communication, as P1 noted, "It's important that the audience knows there can be errors in the podcast [...] just listening to this podcast does not give you enough of an understanding to be super confident." Researchers also suggested incorporating disclaimers, such as P3's recommendation: "This podcast was generated using generative AI... take caution and check your sources." Additionally, P7 further elaborated that a key difference between the AI podcasts and most research papers is that AI podcasts do not go through a peer review process before being published, exacerbating the potential for spreading inaccurate information.

Participants also proposed collaborating with researchers to refine the tool. P8 proposed creating a feedback loop between researchers and developers, stating, "Researchers listening to it and giving feedback to [the developers] could be really productive. Like, 'hey, you completely missed the main point of my paper!' [...] so that [the developers] are aware of where they're falling short."

5 Discussion

Our study explored the reactions of 10 authors of research papers to AI-generated podcasts summarizing their work. In the following section, we describe the key insights we gained from our study, the current limitations of our study, and relevant future work.

5.1 Key Findings

5.1.1 Types of inaccuracies. We found that authors identified a number of issues with the way AI podcasts represented their work, including inappropriate recommendations for applications of their research, the inclusion of relevant yet inaccurate details not addressed by what they wrote in the paper, and the inclusion of extraneous information irrelevant to their research. The insights generated from this study can inform how evaluations of AI-assisted science communication tools are designed in the future.

5.1.2 Authors' Ambivalent Stance Toward AI Podcasts. While authors acknowledge the importance of communicating their work to broader audiences, scientific work is often distorted when translated by people who are not the original researchers or it is summarized into a more concise and accessible message [15]. Indeed, while the authors we interviewed often pointed out hallucinations generated by the podcasts, that did not always deter them from suggesting that the podcasts in their current form could benefit audiences outside their field. Our work shows that while current methods of evaluating features such as the informativeness and faithfulness of AI-assisted science communication tools are useful, they may not provide the complete picture of the benefits and risks of disseminating the outputs of such tools.

5.1.3 Risks of using AI in science communication. The podcasts we played for our participants often adopted an overwhelmingly positive stance towards the participants' papers, which the participants perceived as inauthentic due to the fact the heavy praise was coming from a machine and not a human. The consistent level of praise presented in all of these podcasts hints that AI-assisted science communication tools may not be accurate at judging how research papers may be viewed by the broader research community. On the other hand, the ability to customize LLM outputs with prompting also presents the risk of users being able to generate podcasts that adopt a negative tone toward work that is accepted by the scientific community. Overall, the shortcomings of AI-generated podcasts in critically evaluating research papers within the context of the scientific consensus, paired with the speed at which these podcasts can be generated, presents a unique challenge in preventing misinformation about science.

5.2 Limitations

There are several limitations to our study that we can address in future work. First, while we intended to recruit a broad sample of academics, there are many fields not included in our sample. For example, scholars in the medical field were also absent from our sample, which is noteworthy because many studies of science communication are focused on medical research [6, 18]. In addition to the limitation that many academic fields are absent from our study, our overall sample size is quite small, meaning interviewing more researchers may lead to new insights and a clearer characterization of themes. Similarly, the small sample size hinders our ability to make conclusions about researchers from any given fields and differences in perspectives among fields. Finally, we acknowledge that many of the participants' ideas for interactivity and customization in the podcasts have been explored by previous research [4, 5, 17], and the Notebook LM tool allows users to provide custom instructions when generating podcasts. We opted not to include personalization in our current study design for consistency and simplicity. Nonetheless, presenting the customization feature to the authors could have impacted their views of the AI podcasts, which could be addressed in future work.

6 Conclusion and Future Work

Our work shows that authors offer valuable perspectives into the accuracy and utility of AI-generated podcasts based on their papers. Future work can provide more robust insights by interviewing

a larger number of authors from a wider variety of fields. Similarly, our work can be built on by including authors to co-design AI-assisted science communication tools, iteratively gaining their insights on how to best design safeguards for fighting against misinformation generated by these tools or how to produce more engaging podcasts. Ultimately, our work offers a path forward for designing AI-assisted science communication that is more safe, informative, and accessible.

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